ClipSAT : Egyptian Scholastic Test: EST II Mathematics Level 2 [ANSWER KEY]

Unit 6.01: Permutations, Combinations & Conditional Probability

Q1. If i = sqrt{-1} , which of the following is equivalent to the complex expression frac{7 + 4i}{2 - i} ?

[x] A: 2 + 3i

[ ] B: 2 - 3i

[ ] C: frac{10 + 15i}{3}

[ ] D: frac{18 + 15i}{5}

Solution: To simplify the quotient of two complex numbers, multiply both the numerator and the denominator by the complex conjugate of the denominator, which is 2 + i : frac{7 + 4i}{2 - i} = frac{7 + 4i2 + i}{2 - i2 + i} Expand the numerator using the distributive property FOIL: 7 + 4i2 + i = 14 + 7i + 8i + 4i^2 Since i^2 = -1 : 14 + 15i + 4-1 = 14 - 4 + 15i = 10 + 15i Expand the denominator using the difference of squares: 2 - i2 + i = 2^2 - i^2 = 4 - -1 = 5 Now, divide each term in the numerator by the denominator: frac{10 + 15i}{5} = frac{10}{5} + frac{15i}{5} = 2 + 3i Common Misconceptions: - Choice B occurs from an arithmetic sign error in the imaginary term. - Choice C occurs if the student mistakenly calculates the denominator as 2^2 - 1^2 = 3 by forgetting that -i^2 = --1 = +1 . - Choice D occurs if the student takes i^2 = 1 , yielding 14 + 4 = 18 in the numerator.

Q2. The graph of the function y = fx is shown below. What is the value of f1 ?

[x] A: 0

[ ] B: 2

[ ] C: -1

[ ] D: 3

Solution: The value of f1 corresponds to the y-coordinate of the point on the graph where the x-coordinate is 1. Looking at the provided graph, the point 1, 0 is marked. Therefore, f1 = 0 .

Q3. Consider the function fx = x^2 - 4x + 3 . What are the x-intercepts of the graph of this function?

[x] A: 1, 0 and 3, 0

[ ] B: 0, 1 and 0, 3

[ ] C: -1, 0 and -3, 0

[ ] D: 1, 0 and -3, 0

Solution: The x-intercepts of a function occur where the function's value is zero, i.e., fx = 0 . nWe need to solve the quadratic equation x^2 - 4x + 3 = 0 . nThis equation can be factored as x-1x-3 = 0 . nTherefore, the solutions are x=1 and x=3 . nThis means the x-intercepts are 1, 0 and 3, 0 .

Q4. A vector mathbf{v} is represented by the directed line segment from point P1, 2 to point Q4, 6 . What are the components of vector mathbf{v} ?

[x] A: [3, 4]

[ ] B: [5, 8]

[ ] C: [-3, -4]

[ ] D: [4, 3]

Solution: The components of a vector mathbf{v} represented by a directed line segment from point Px_1, y_1 to point Qx_2, y_2 are given by mathbf{v} = [x_2 - x_1, y_2 - y_1] . In this case, P = 1, 2 and Q = 4, 6 . Therefore, the components of mathbf{v} are [4 - 1, 6 - 2] = [3, 4] .

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Q5. A bag contains 12 red marbles and 8 blue marbles. If one marble is drawn at random, what is the probability that the marble is red?

[x] A: frac{3}{5}
[ ] B: frac{2}{5}
[ ] C: frac{1}{2}
[ ] D: frac{3}{4}

Solution: The total number of marbles in the bag is the sum of red and blue marbles: 12 + 8 = 20 marbles. The number of favorable outcomes drawing a red marble is 12. The probability of an event is calculated as the ratio of the number of favorable outcomes to the total number of possible outcomes. Therefore, the probability of drawing a red marble is Ptext{red} = frac{text{Number of red marbles}}{text{Total number of marbles}} = frac{12}{20} . This fraction can be simplified by dividing both the numerator and the denominator by their greatest common divisor, which is 4: frac{12 div 4}{20 div 4} = frac{3}{5} . Thus, the probability of drawing a red marble is frac{3}{5} .

Q6. Consider an arithmetic sequence where the first term is a_1 = 5 and the common difference is d = 3 . What is the value of the 4th term, a_4 ?

[x] A: 14
[ ] B: 17
[ ] C: 8
[ ] D: 11

Solution: The formula for the nth term of an arithmetic sequence is given by a_n = a_1 + n-1d . In this problem, we are given a_1 = 5 and d = 3 . We need to find the 4th term, so n = 4 . Plugging these values into the formula, we get: a_4 = 5 + 4-1 times 3 a_4 = 5 + 3 times 3 a_4 = 5 + 9 a_4 = 14 . Therefore, the 4th term of the sequence is 14.

Q7. A geometric sequence has a_1=6 and common ratio r=2 . Find a_{5} .

[x] A: 96
[ ] B: 192
[ ] C: 98
[ ] D: -96

Solution: Applying Sequences & Series principles, a_{5}=a_1r^{n-1}=62^{4}=96 . Therefore, the verified correct option is 96.

Q8. The graph of the function y = fx = ax^2 + bx + c is shown below. The graph passes through the points -1, 5 , 1, 1 , and 3, 5 . Determine the value of b .

[ ] A: 0
[x] B: -2
[ ] C: 2
[ ] D: 1

Solution: The general form of the quadratic function is y = ax^2 + bx + c . We are given three points that lie on the graph: -1, 5 , 1, 1 , and 3, 5 . Substituting these points into the equation, we get a system of three linear equations with three unknowns a, b, and c: 1. For -1, 5 : a-1^2 + b-1 + c = 5 Rightarrow a - b + c = 5 2. For 1, 1 : a1^2 + b1 + c = 1 Rightarrow a + b + c = 1 3. For 3, 5 : a3^2 + b3 + c = 5 Rightarrow 9a + 3b + c = 5 We can solve this system of equations. Let's subtract equation 1 from equation 2: a + b + c - a - b + c = 1 - 5 2b = -4 b = -2 Alternatively, notice that the y-values for x = -1 and x = 3 are the same both are 5. This indicates that the axis of symmetry of the parabola is exactly in the middle of x = -1 and x = 3 . The x-coordinate of the vertex axis of symmetry is given by -frac{b}{2a} . Axis of symmetry: x = frac{-1 + 3}{2} = frac{2}{2} = 1 So, -frac{b}{2a} = 1 , which implies b = -2a . Now, substitute this into equation 2: a + b + c = 1 a + -2a + c = 1 -a + c = 1 Rightarrow c = a + 1 Substitute b = -2a and c = a + 1 into equation 1: a - -2a + a + 1 = 5 a + 2a + a + 1 = 5 4a + 1 = 5 4a = 4 a = 1 Now we can find b using b = -2a : b = -21 b = -2 To confirm, we can also find c : c = a + 1 = 1 + 1 = 2 . The function is y = x^2 - 2x + 2 . Let's check the points: For -1, 5 : -1^2 - 2-1 + 2 = 1 + 2 + 2 = 5 Correct For 1, 1 : 1^2 - 21 + 2 = 1 - 2 + 2 = 1 Correct For 3, 5 : 3^2 - 23 + 2 = 9 - 6 + 2 = 5 Correct The value of b is -2.

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